Performance Testing

Date	27 June 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing (20 concurrent virtual users, ramped up at 5 users/sec)
Target Module / API	/predict (crop recommendation endpoint) and / (home page / input form)
Test Environment	Cloud / Hosted Server – live production deployment on Vercel (https://opticrop-indol.vercel.app)
Test Date	4 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Farmers submitting soil & climate data to /predict and receiving a crop recommendation	20	60	All requests succeed; avg response < 2s
2	Farmers/visitors loading the home page (input form) at /	20	60	Page loads successfully with no errors
3	Combined load: mixed GET / and POST /predict traffic (as a real user session would	20	60	System remains stable with 0% error rate under mixed load
4	Repeat run after idle period (to observe cold-start behavior)	20	60	First request(s) may be slower (cold start); subsequent requests fast

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.20 s (/predict); 1.68 s (/)	Pass	Average well within target for both endpoints
2	Response Time (Max)	< 5 seconds	11.29 s (/predict); 15.13 s (/)	Fail	Max spikes are consistent with Vercel serverless cold starts, not steady-state slowness
3	Throughput (Req/sec)		2.77 req/s (/predict); 5.74 req/s (aggregated)	N/A	No target was set for this metric; reported for reference
4	Error Rate	< 1%	0% (0 failures out of 146 requests to /predict)	Pass	No failed requests during the 60-second test
5	CPU Utilization	< 80%	Not measured	N/A	Vercel serverless functions do not expose CPU metrics to the developer dashboard for this plan tier
6	Memory Utilization	< 80%	Not measured	N/A	Not available from Vercel's free-tier monitoring; would require upgrading the plan or adding custom instrumentation

Step 4: Observations & Analysis

Key Findings:
Under a sustained load of 20 concurrent virtual users over 60 seconds, OptiCrop's /predict endpoint handled 146 requests with a 0% error rate. The median response time was 440ms and the average was 1.20s, comfortably meeting the <2s target. However, maximum response times reached 11.29s for /predict and 15.13s for the home page, well above the 5s target.

Bottlenecks Identified:
The large gap between median (~440ms) and maximum (11-15s) response times is characteristic of Vercel serverless cold starts: when the function hasn't been invoked recently, it must reinitialize the Python runtime and reload the pickled model (crop_model.pkl, label_encoder.pkl) from disk before it can serve a request. This adds several seconds of one-time latency rather than reflecting steady-state performance.

Optimization Steps Taken:
No optimizations have been applied yet. Potential next steps include: scheduling periodic "keep-warm" pings to the endpoint to reduce cold starts, reducing the size of the model/imports to speed up cold-start initialization, or moving to an always-on server if consistently low latency becomes a requirement.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Type	Name	Request C	Failure Co	Median Re	Average R	Min Respo	Max Respc	Average C	Requests/	Failures/s	50%	66%	75%	80%	90%	95%	98%	99%	99.90%	99.99%	100%
GET	/	156	0	440	1675.626	258.2277	15128.12	3334	2.963593	0	450	680	1100	1400	3600	11000	13000	14000	15000	15000	15000
POST	/predict	146	0	440	1202.888	284.4931	11293.52	2955.507	2.773619	0	460	690	1100	1300	2300	4000	10000	11000	11000	11000	11000
	Aggregate	302	0	440	1447.084	258.2277	15128.12	3151.02	5.737211	0	450	690	1100	1300	3100	10000	11000	13000	15000	15000	15000